Python Development

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assignment code (day 5)

Question – Smart Inventory Management system

Code –

```
inventory = {}
categories = set()
def add_product():
   pid = input("Enter Product ID: ")
   if pid in inventory:
      print("Product already exists")
      return
   name = input("Enter Product Name: ")
   category = input("Enter Category: ")
   qty = int(input("Enter Quantity: "))
   price = float(input("Enter Price: "))
   supplier = input("Enter Supplier: ")
   inventory[pid] = {
      "name": name,
      "category": category,
      "qty": qty,
      "price": price,
      "supplier": supplier
   }
   categories.add(category)
```

```
    print("Product Added Successfully")
def update_inventory():
    pid = input("Enter Product ID: ")
    if pid in inventory:
        inventory[pid]["qty"] = int(input("Enter New Quantity: "))
        inventory[pid]["price"] = float(input("Enter New Price: "))
        print("Inventory Updated")
    else:
        print("Product Not Found")
def search_product():
    key = input("Enter Product ID or Name: ").lower()
    found = []
    for pid, details in inventory.items():
        if key == pid.lower() or key == details["name"].lower():
            found.append((pid, details))
    if found:
        for item in found:
            print(item)
    else:
        print("Product Not Found")
def display_inventory():
    if not inventory:
        print("Inventory Empty")
        return
    print("\nID\tName\tCategory\tQty\tPrice\tSupplier")
```

```
    for pid, details in inventory.items():
        print(
            pid,
            details["name"],
            details["category"],
            details["qty"],
            details["price"],
            details["supplier"],
            sep="\t"
        )

def low_stock_alert():
    threshold = 10

    for pid, details in inventory.items():
        if details["qty"] < threshold:
            print(pid, details["name"], "- Low Stock")

def out_of_stock_alert():
    for pid, details in inventory.items():
        if details["qty"] == 0:
            print(pid, details["name"], "- Out Of Stock")

def category_management():
    print("Categories:")
    for category in categories:
```

```
        print(category)

def inventory_report():
    total_items = 0
    total_value = 0

    report = {}

    for details in inventory.values():
        total_items += details["qty"]
        total_value += details["qty"] * details["price"]

        category = details["category"]

        if category not in report:
            report[category] = 0

        report[category] += details["qty"]

    snapshot = (total_items, total_value)

    print("Total Items:", snapshot[0])
    print("Total Inventory Value:", snapshot[1])

    print("Category Wise Summary:")
    for category, qty in report.items():
```

```
        print(category, ":", qty)

def delete_product():
    pid = input("Enter Product ID: ")

    if pid in inventory:
        del inventory[pid]
        print("Product Deleted")
    else:
        print("Product Not Found")

while True:
    print("\nSMART INVENTORY MANAGEMENT SYSTEM")
    print("1. Add Product")
    print("2. Update Inventory")
    print("3. Search Product")
    print("4. Display Inventory")
    print("5. Low Stock Alert")
    print("6. Out Of Stock Alert")
    print("7. Category Management")
    print("8. Inventory Report")
    print("9. Delete Product")
    print("10. Exit")

    choice = input("Enter Choice: ")
```

```
    if choice == "1":
        add_product()
    elif choice == "2":
        update_inventory()
    elif choice == "3":
        search_product()
    elif choice == "4":
        display_inventory()
    elif choice == "5":
        low_stock_alert()
    elif choice == "6":
        out_of_stock_alert()
    elif choice == "7":
        category_management()
    elif choice == "8":
        inventory_report()
    elif choice == "9":
        delete_product()
    elif choice == "10":
        print("Exiting...")
        break
    else:
        print("Invalid Choice")
```

Output –

SMART INVENTORY MANAGEMENT SYSTEM

1. Add Product

2. Update Inventory

3. Search Product

4. Display Inventory

5. Low Stock Alert

6. Out Of Stock Alert

7. Category Management

8. Inventory Report

9. Delete Product

10. Exit

Enter Choice: 1

Enter Product ID: 101

Enter Product Name: Pen

Enter Category: Stationery

Enter Quantity: 4

Enter Price: 20

Enter Supplier: abc

Product Added Successfully

SMART INVENTORY MANAGEMENT SYSTEM

1. Add Product

2. Update Inventory

3. Search Product

4. Display Inventory

5. Low Stock Alert

6. Out Of Stock Alert

7. Category Management

8. Inventory Report

9. Delete Product

10. Exit

Enter Choice: 3

Enter Product ID or Name: 101

('101', {'name': 'Pen', 'category': 'Stationery', 'qty': 4, 'price': 20.0, 'supplier': 'abc'})

SMART INVENTORY MANAGEMENT SYSTEM

1. Add Product

2. Update Inventory

3. Search Product

4. Display Inventory

5. Low Stock Alert

6. Out Of Stock Alert

7. Category Management

8. Inventory Report

9. Delete Product

10. Exit

Enter Choice: 7

Categories:

Stationery

SMART INVENTORY MANAGEMENT SYSTEM

1. Add Product

2. Update Inventory

3. Search Product

4. Display Inventory

5. Low Stock Alert

6. Out Of Stock Alert

7. Category Management

8. Inventory Report

9. Delete Product

10. Exit

Enter Choice: 8

Total Items: 4

Total Inventory Value: 80.0

Category Wise Summary:

Stationery : 4

```
SMART INVENTORY MANAGEMENT SYSTEM
1. Add Product
2. Update Inventory
3. Search Product
4. Display Inventory
5. Low Stock Alert
6. Out Of Stock Alert
7. Category Management
8. Inventory Report
9. Delete Product
10. Exit
Enter Choice: 1
Enter Product ID: 101
Enter Product Name: Pen
Enter Category: Stationery
Enter Quantity: 4
Enter Price: 20
Enter Supplier: abc
Product Added Successfully

SMART INVENTORY MANAGEMENT SYSTEM
1. Add Product
2. Update Inventory
3. Search Product
4. Display Inventory
5. Low Stock Alert
6. Out Of Stock Alert
7. Category Management
8. Inventory Report
9. Delete Product
10. Exit
Enter Choice: 3
Enter Product ID or Name: 101
('101', {'name': 'Pen', 'category': 'Stationery', 'qty': 4, 'price': 20.0, 'supplier': 'abc'})
```

```
SMART INVENTORY MANAGEMENT SYSTEM
1. Add Product
2. Update Inventory
3. Search Product
4. Display Inventory
5. Low Stock Alert
6. Out Of Stock Alert
7. Category Management
8. Inventory Report
9. Delete Product
10. Exit
Enter Choice: 7
Categories:
Stationery

SMART INVENTORY MANAGEMENT SYSTEM
1. Add Product
2. Update Inventory
3. Search Product
4. Display Inventory
5. Low Stock Alert
6. Out Of Stock Alert
7. Category Management
8. Inventory Report
9. Delete Product
10. Exit
Enter Choice: 8
Total Items: 4
Total Inventory Value: 80.0
Category Wise Summary:
Stationery : 4
```